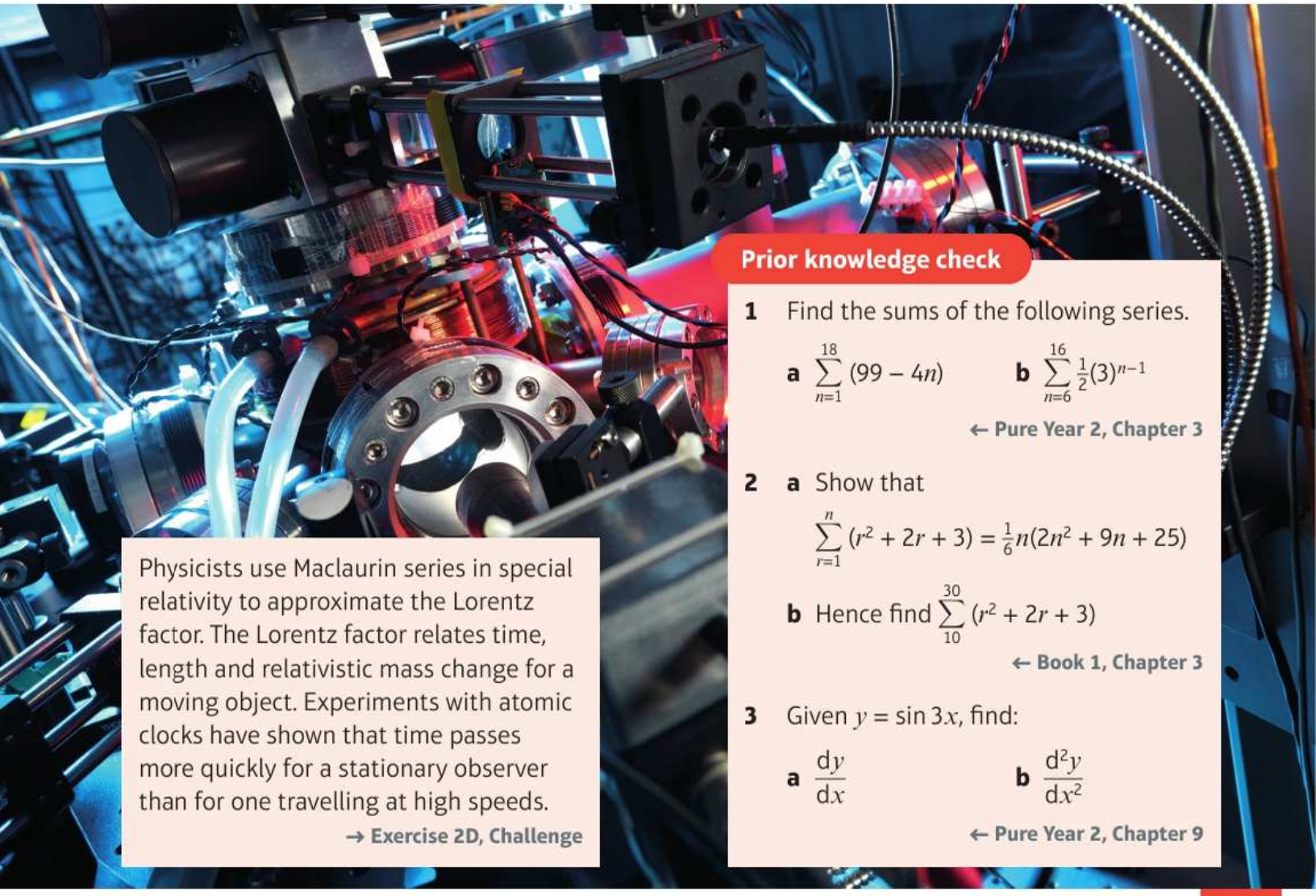
Series

2

Objectives

After completing this chapter you should be able to:

- Understand and use the method of differences to sum finite series → pages 32–37
- Find and use higher derivatives of functions → pages 38–39
- Know how to express functions as an infinite series in ascending powers using Maclaurin series expansion → pages 40–44
- Be able to find the series expansions of compound functions → pages 44–48



Physicists use Maclaurin series in special relativity to approximate the Lorentz factor. The Lorentz factor relates time, length and relativistic mass change for a moving object. Experiments with atomic clocks have shown that time passes more quickly for a stationary observer than for one travelling at high speeds.

→ Exercise 2D, Challenge

Prior knowledge check

1 Find the sums of the following series.

a $\sum_{n=1}^{18} (99 - 4n)$ **b** $\sum_{n=6}^{16} \frac{1}{2}(3)^{n-1}$

← Pure Year 2, Chapter 3

2 **a** Show that

$$\sum_{r=1}^{n} (r^2 + 2r + 3) = \frac{1}{6}n(2n^2 + 9n + 25)$$

b Hence find $\sum_{10}^{30} (r^2 + 2r + 3)$

← Book 1, Chapter 3

3 Given $y = \sin 3x$, find:

a $\frac{dy}{dx}$ **b** $\frac{d^2y}{dx^2}$

← Pure Year 2, Chapter 9

2.1 The method of differences

You can use the method of differences to find the sum of a finite series.

- **If the general term, u_r, of a series can be expressed in the form**

$$\mathbf{f(r) - f(r+1)}$$

then $\displaystyle\sum_{r=1}^{n} u_r = \sum_{r=1}^{n} (\mathbf{f(r) - f(r+1)})$

You can also start with u_r written in the form $f(r+1) - f(r)$. After adding and cancelling, you get $\displaystyle\sum_{r=1}^{n} u_r = f(n+1) - f(1)$

so $u_1 = \mathbf{f(1) - f(2)}$

$u_2 = \mathbf{f(2) - f(3)}$

$u_3 = \mathbf{f(3) - f(4)}$

$\vdots$

$u_n = \mathbf{f(n) - f(n+1)}$

$u_1 + u_2 = f(1) - f(2) + f(2) - f(3)$
$= f(1) - f(3)$

Then adding $\displaystyle\sum_{r=1}^{n} u_r = \mathbf{f(1) - f(n+1)}$

The f(2) terms cancel.

By summing $u_1 + u_2 + \ldots + u_n$ all terms cancel except the very first term, f(1), and the very last term, $f(n+1)$.

Example 1

a Show that $4r^3 \equiv r^2(r+1)^2 - (r-1)^2r^2$

b Hence prove, by the method of differences, that

$$\sum_{r=1}^{n} r^3 = \tfrac{1}{4}n^2(n+1)^2$$

a $r^2(r+1)^2 - (r-1)^2r^2$

Start with the RHS.

$\equiv r^2(r^2 + 2r + 1) - (r^2 - 2r + 1)r^2$

$\equiv r^4 + 2r^3 + r^2 - r^4 + 2r^3 - r^2$

Expand and simplify the brackets.

$\equiv 4r^3$

b Consider $\displaystyle\sum_{r=1}^{n} (r^2(r+1)^2 - (r-1)^2r^2)$

Let $r = 1$: ~~$1^2(2)^2$~~ $- (0)^2 1^2$

$r = 2$: ~~$2^2(3)^2$~~ $-$ ~~$(1)^2 2^2$~~

$r = 3$: ~~$3^3(4)^2$~~ $-$ ~~$(2)^2 2^3$~~

$\vdots$

$r = n$: $n^2(n+1)^2 -$ ~~$(n-1)^2 n^2$~~

All the terms cancel except the first and last.

Sum of terms $= n^2(n+1)^2$

Then $4\displaystyle\sum_{r=1}^{n} r^3 = n^2(n+1)^2$

So $\displaystyle\sum_{r=1}^{n} r^3 = \tfrac{1}{4}n^2(n+1)^2$

Watch out When using the method of differences, be sure to write out enough terms to make it clear which terms cancel. When you cancel terms, make sure that they can still be clearly read. You could cross them out in pencil.

The same result could be proved by mathematical induction. ← Book 1, Chapter 8

Example 2

Verify that $\frac{1}{r(r+1)} \equiv \frac{1}{r} - \frac{1}{r+1}$ and hence find $\sum_{r=1}^{n} \frac{1}{r(r+1)}$ using the method of differences.

$\frac{1}{r} - \frac{1}{r+1} \equiv \frac{r+1-r}{r(r+1)}$ — Write as a single fraction.

$\equiv \frac{1}{r(r+1)}$ — Simplify.

$\sum_{r=1}^{n} \frac{1}{r(r+1)} \equiv \sum_{r=1}^{n} \left(\frac{1}{r} - \frac{1}{r+1}\right)$

Let $r = 1$: $\frac{1}{1} - \frac{1}{2}$

$r = 2$: $\frac{1}{2} - \frac{1}{3}$

$r = 3$: $\frac{1}{3} - \frac{1}{4}$

$\vdots$

$r = n$: $\frac{1}{n} - \frac{1}{n+1}$

All terms cancel except the first and last.

So $\sum_{r=1}^{n} \frac{1}{r(r+1)} = 1 - \frac{1}{n+1}$

$= \frac{n+1-1}{n+1}$ — Put over a common denominator.

$= \frac{n}{n+1}$

Example 3

Find $\sum_{r=1}^{n} \frac{1}{4r^2 - 1}$ using the method of differences.

$\frac{1}{4r^2 - 1} \equiv \frac{1}{(2r+1)(2r-1)}$ — Use the difference of two squares to factorise the denominator.

$\frac{1}{(2r+1)(2r-1)} \equiv \frac{A}{2r+1} + \frac{B}{2r-1}$ — Split the fraction into partial fractions.

← Pure Year 2, Chapter 1

$\equiv \frac{A(2r-1) + B(2r+1)}{(2r+1)(2r-1)}$ — Add the fractions.

so $1 \equiv A(2r-1) + B(2r+1)$ — Set numerators of both sides equal to each other.

Let $r = \frac{1}{2}$: $1 = 0 + B \times 2$

$B = \frac{1}{2}$

Let $r = -\frac{1}{2}$: $1 = A \times -2$

$A = -\frac{1}{2}$

So $\frac{1}{4r^2 - 1} \equiv \frac{-\frac{1}{2}}{2r + 1} + \frac{\frac{1}{2}}{2r - 1}$

$\equiv \frac{1}{2}\left(\frac{1}{2r - 1} - \frac{1}{2r + 1}\right)$

$\Rightarrow \sum_{r=1}^{n} \frac{1}{4r^2 - 1} = \frac{1}{2}\sum_{r=1}^{n}\left(\frac{1}{2r - 1} - \frac{1}{2r + 1}\right)$

Let $r = 1$: $\frac{1}{1} - \frac{1}{3}$

$r = 2$: $\frac{1}{3} - \frac{1}{5}$

$r = 3$: $\frac{1}{5} - \frac{1}{7}$

$\vdots$

$r = n$: $\frac{1}{2n - 1} - \frac{1}{2n + 1}$

So $\sum_{r=1}^{n} \frac{1}{4r^2 - 1} = \frac{1}{2}\left(1 - \frac{1}{2n + 1}\right)$

$= \frac{1}{2}\left(\frac{2n + 1 - 1}{2n + 1}\right)$

$= \frac{n}{2n + 1}$

Put values of r in to find A and B.

All terms cancel except the first and last.
Substitute the values of r into $\frac{1}{2r - 1} - \frac{1}{2r + 1}$ only. The $\frac{1}{2}$ is only required later.

If the general term of the series is given in the form $f(r) - f(r + 2)$, you need to adapt the method of differences to consider the terms $f(1)$, $f(2)$, $f(n + 1)$ and $f(n + 2)$.

Example 4

a Express $\frac{2}{(r + 1)(r + 3)}$ in partial fractions.

b Hence prove by the method of differences that

$$\sum_{r=1}^{n} \frac{2}{(r + 1)(r + 3)} = \frac{n(an + b)}{6(n + 2)(n + 3)}$$

where a and b are constants to be found.

c Find the value of $\sum_{r=21}^{30} \frac{2}{(r + 1)(r + 3)}$ to 5 decimal places.

a $\frac{2}{(r+1)(r+3)} \equiv \frac{A}{r+1} + \frac{B}{r+3}$

Split into partial fractions.

$\equiv \frac{A(r+3) + B(r+1)}{(r+1)(r+3)}$

Add the fractions.

$\Rightarrow \quad 2 \equiv A(r+3) + B(r+1)$

Compare numerators.

Let $r = -3$: $2 = -2B \Rightarrow B = -1$

Let $r = -1$: $2 = 2A \Rightarrow A = 1$

Therefore $\frac{2}{(r+1)(r+3)} \equiv \frac{1}{r+1} - \frac{1}{r+3}$

b Using the method of differences,

when $r = 1$: $\frac{1}{2} - \cancel{\frac{1}{4}}$

$r = 2$: $\frac{1}{3} - \cancel{\frac{1}{5}}$

$r = 3$: $\cancel{\frac{1}{4}} - \cancel{\frac{1}{6}}$

$\vdots$

$r = n - 1$: $\cancel{\frac{1}{n}} - \frac{1}{n+2}$

$r = n$: $\frac{1}{\cancel{n+1}} - \frac{1}{n+3}$

Cancel terms.

Problem-solving

$\sum_{r=1}^{n}(f(r) - f(r+2)) =$
$f(1) + f(2) - f(n+1) - f(n+2)$

So $\sum_{r=1}^{n} \frac{2}{(r+1)(r+3)} = \frac{5}{6} - \frac{1}{n+2} - \frac{1}{n+3}$

Put these four terms over a common denominator.

$= \frac{5(n+2)(n+3) - 6(n+3) - 6(n+2)}{6(n+2)(n+3)}$

$= \frac{5n^2 + 25n + 30 - 6n - 18 - 6n - 12}{6(n+2)(n+3)}$

$= \frac{5n^2 + 13n}{6(n+2)(n+3)}$

$= \frac{n(5n+13)}{6(n+2)(n+3)}$

Factorise.

So $a = 5$ and $b = 13$.

c $\sum_{r=21}^{30} \frac{2}{(r+1)(r+3)} = \sum_{r=1}^{30} \frac{2}{(r+1)(r+3)} - \sum_{r=1}^{20} \frac{2}{(r+1)(r+3)}$

Subtract $\sum_{r=1}^{20}$ from $\sum_{r=1}^{30}$.

$= \frac{30(5 \times 30 + 13)}{6(30+2)(30+3)} - \frac{20(5 \times 20 + 13)}{6(20+2)(20+3)}$

$= \frac{815}{1056} - \frac{565}{759}$

Evaluate.

$= \frac{665}{24\,288} = 0.02738$ to 5 d.p.

Give answer to 5 d.p.

a $\frac{2}{(r+1)(r+3)} \equiv \frac{A}{r+1} + \frac{B}{r+3}$

Split into partial fractions.

$\equiv \frac{A(r+3) + B(r+1)}{(r+1)(r+3)}$

Add the fractions.

$\Rightarrow \quad 2 \equiv A(r+3) + B(r+1)$

Compare numerators.

Let $r = -3$: $2 = -2B \Rightarrow B = -1$

Let $r = -1$: $2 = 2A \Rightarrow A = 1$

Therefore $\frac{2}{(r+1)(r+3)} \equiv \frac{1}{r+1} - \frac{1}{r+3}$

b Using the method of differences,

when $r = 1$: $\frac{1}{2} - \cancel{\frac{1}{4}}$

$r = 2$: $\frac{1}{3} - \cancel{\frac{1}{5}}$

$r = 3$: $\cancel{\frac{1}{4}} - \cancel{\frac{1}{6}}$

$\vdots$

$r = n - 1$: $\cancel{\frac{1}{n}} - \frac{1}{n+2}$

$r = n$: $\frac{1}{\cancel{n+1}} - \frac{1}{n+3}$

Cancel terms.

Problem-solving

$\sum_{r=1}^{n}(f(r) - f(r+2)) =$
$f(1) + f(2) - f(n+1) - f(n+2)$

So $\sum_{r=1}^{n} \frac{2}{(r+1)(r+3)} = \frac{5}{6} - \frac{1}{n+2} - \frac{1}{n+3}$

Put these four terms over a common denominator.

$= \frac{5(n+2)(n+3) - 6(n+3) - 6(n+2)}{6(n+2)(n+3)}$

$= \frac{5n^2 + 25n + 30 - 6n - 18 - 6n - 12}{6(n+2)(n+3)}$

$= \frac{5n^2 + 13n}{6(n+2)(n+3)}$

$= \frac{n(5n+13)}{6(n+2)(n+3)}$

Factorise.

So $a = 5$ and $b = 13$.

c $\sum_{r=21}^{30} \frac{2}{(r+1)(r+3)} = \sum_{r=1}^{30} \frac{2}{(r+1)(r+3)} - \sum_{r=1}^{20} \frac{2}{(r+1)(r+3)}$

Subtract $\sum_{r=1}^{20}$ from $\sum_{r=1}^{30}$.

$= \frac{30(5 \times 30 + 13)}{6(30+2)(30+3)} - \frac{20(5 \times 20 + 13)}{6(20+2)(20+3)}$

$= \frac{815}{1056} - \frac{565}{759}$

Evaluate.

$= \frac{665}{24\,288} = 0.02738$ to 5 d.p.

Give answer to 5 d.p.

Exercise 2A

1 a Show that $r \equiv \frac{1}{2}(r(r+1) - r(r-1))$.

b Hence show that $\sum_{r=1}^{n} r = \frac{n}{2}(n+1)$ using the method of differences.

E **2** Given $\frac{1}{r(r+1)(r+2)} \equiv \frac{1}{2r(r+1)} - \frac{1}{2(r+1)(r+2)}$

find $\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)}$ using the method of differences. **(5 marks)**

E/P **3 a** Express $\frac{1}{r(r+2)}$ in partial fractions. **(1 mark)**

b Hence find the sum of the series $\sum_{r=1}^{n} \frac{1}{r(r+2)}$ using the method of differences. **(5 marks)**

E **4 a** Express $\frac{1}{(r+2)(r+3)}$ in partial fractions. **(1 mark)**

b Hence find the sum of the series $\sum_{r=1}^{n} \frac{1}{(r+2)(r+3)}$ using the method of differences. **(5 marks)**

E/P **5 a** Show that $\frac{r}{(r+1)!} \equiv \frac{1}{r!} - \frac{1}{(r+1)!}$ **(2 marks)**

b Hence find $\sum_{r=1}^{n} \frac{r}{(r+1)!}$ **(5 marks)**

E **6** Given that $\frac{2r+1}{r^2(r+1)^2} \equiv \frac{1}{r^2} - \frac{1}{(r+1)^2}$, find $\sum_{r=1}^{n} \frac{2r+1}{r^2(r+1)^2}$ **(6 marks)**

P **7 a** Use the method of differences to prove that $\sum_{r=1}^{n} \frac{1}{2r+32r+5} = \frac{n}{an+b}$, where a and b are constants to be found.

b Prove your result from part **a** using mathematical induction.

E/P **8** Prove that $\sum_{r=1}^{n} \frac{8}{3r-23r+4} = \frac{n(an+b)}{(3n+1)(3n+4)}$, where a and b are constants to be found. **(6 marks)**

Hint This question can be answered using either the method of differences or proof by induction. In the exam, either method would be acceptable. If you use proof by induction, you will need to substitute values of n to find the values of a and b.

E/P **9** Prove that $\sum_{r=1}^{n} (r+1)^2 - (r-1)^2 = an(n+1)$, where a is a constant to be found. **(4 marks)**

E/P **10** **a** Prove that $\sum_{r=1}^{n} \frac{3}{(3r+1)(3r+4)} = \frac{an}{bn+c}$, where a, b and c are constants to be found. **(5 marks)**

b Hence, or otherwise, show that $\sum_{r=n}^{2n} \frac{3}{3r+13r+4} = \frac{3(n+1)}{2(3n+1)(3n+2)}$ **(4 marks)**

E/P **11** Robin claims that $\sum_{r=1}^{n} \frac{2r+1}{r(r+1)} = 1 - \frac{1}{n+1}$

His workings are shown below. Explain the error that he has made.

Using partial fractions:

$$\frac{2r+1}{r(r+1)} \equiv \frac{A}{r} + \frac{B}{r+1}$$

Therefore $2r + 1 \equiv A(r + 1) + Br$

So $A = 1$ and $B = 1$.

Using the method of differences,

$f(1) = 1 + \frac{1}{2}$

$f(2) = \frac{1}{2} + \frac{1}{3}$

$f(3) = \frac{1}{3} + \frac{1}{4}$

$\vdots$

$f(n-1) = \frac{1}{n-1} + \frac{1}{n}$

$f(n) = \frac{1}{n} + \frac{1}{n+1}$

Summing the differences: $\sum_{r=1}^{n} \frac{2r+1}{r(r+1)} = 1 - \frac{1}{n+1}$

(2 marks)

E/P **12** Show that $\frac{1}{1\times 3} + \frac{1}{2\times 4} + \frac{1}{3\times 5} + \ldots + \frac{1}{n(n+2)} = \frac{3}{4} - \frac{an+b}{2(n+1)(n+2)}$, where a and b are constants to be found. **(6 marks)**

E/P **13** **a** Express $\frac{4}{(2r+1)(2r+5)}$ in partial fractions. **(3 marks)**

b Find the value of $\sum_{r=16}^{25} \frac{4}{(2r+1)(2r+5)}$ to 4 decimal places. **(5 marks)**

Challenge

a Given that $\sum_{r=1}^{30} \ln\left(1 + \frac{1}{r+2}\right) = \ln k$, where k is an integer, find k.

b Given that $\sum_{r=1}^{n} \frac{18}{r(r+3)} = \frac{n(an^2+bn+c)}{(n+1)(n+3)(n+3)}$, find a, b and c.

2.2 Higher derivatives

You need to be able to find third, and higher, derivatives of given functions.
You already know how to find first and second derivatives.

If $y = f(x)$, the first derivative of $f(x)$ is given by $\frac{dy}{dx} = f'(x)$, and the second derivative of $f(x)$ is given by $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = f''(x)$.

Similarly, the third derivative is given by $\frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = f'''(x)$, and so on.

You can find the nth derivative of $f(x)$ by differentiating n times with respect to x.

Notation The nth derivative of $y = f(x)$ is written as $\frac{d^ny}{dx^n} = f^{(n)}(x)$.

Example 5

Given that $y = \ln(1 - x)$, find the value of $\frac{d^3y}{dx^3}$ when $x = \frac{1}{2}$

$\frac{dy}{dx} = \frac{1}{1-x}\frac{d}{dx}(1-x) = \frac{1}{1-x} \times (-1) = -\frac{1}{1-x}$ — Use the chain rule. ← Pure Year 2, Section 9.3

$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}(-(1-x)^{-1}) = \frac{1}{(1-x)^2} \times (-1) = -\frac{1}{(1-x)^2}$

$\frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d}{dx}(-(1-x)^{-2}) = \frac{2}{(1-x)^3} \times (-1) = -\frac{2}{(1-x)^3}$

So when $x = \frac{1}{2}$, $\frac{d^3y}{dx^3} = \frac{-2}{\left(1-\frac{1}{2}\right)^3} = -16$ — Substitute $x = \frac{1}{2}$

Example 6

$f(x) = e^{x^2}$

a Show that $f'(x) = 2xf(x)$.

b By differentiating the result in part **a** twice more with respect to x, show that:
i $f''(x) = 2f(x) + 2xf'(x)$ **ii** $f'''(x) = 2xf''(x) + 4f'(x)$

c Deduce the values of $f'(0)$, $f''(0)$, and $f'''(0)$.

a $f'(x) = e^{x^2}\frac{d}{dx}(x^2) = 2xe^{x^2}$ — If $f(x) = e^u$, then $f'(x) = e^u\frac{du}{dx}$

$= 2xf(x)$ — $f(x) = e^{x^2}$

b i $f''(x) = 2f(x) + 2xf'(x)$ — Use the product rule. ← Pure Year 2, Section 9.4

ii $f'''(x) = 2f'(x) + (2xf''(x) + 2f'(x))$ — Differentiate again.

$= 2xf''(x) + 4f'(x)$

c $f(0) = e^0 = 1$

$f'(0) = 2 \times 0 \times e^0 = 0$

$f''(0) = 2f(0) + 2 \times 0 \times f'(0)$ — Substitute $x = 0$ into $f''(x)$.

$= 2f(0) = 2$

$f'''(0) = 2 \times 0 \times f''(0) + 4f'(0)$ — Substitute $x = 0$ into $f'''(x)$.

$= 4f'(0) = 0$

Exercise 2B

1 For each of the following functions, $f(x)$, find $f'(x)$, $f''(x)$, $f'''(x)$ and $f^{(n)}(x)$.

a e^{2x} **b** $(1 + x)^n$ **c** xe^x **d** $\ln(1 + x)$

(P) **2 a** Given that $y = e^{2+3x}$, find an expression, in terms of y, for $\frac{d^n y}{dx^n}$

b Hence evaluate $\frac{d^6 y}{dx^6}$ when $x = \ln\left(\frac{1}{9}\right)$.

3 Given that $y = \sin^2 3x$,

a show that $\frac{dy}{dx} = 3 \sin 6x$ **b** find expressions for $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$ and $\frac{d^4y}{dx^4}$

c Hence evaluate $\frac{d^4 y}{dx^4}$ when $x = \frac{\pi}{6}$

4 $f(x) = x^2e^{-x}$.

a Show that $f'''(x) = (6x - 6 - x^2)e^{-x}$. **b** Show that $f''''(2) = 0$.

5 Given that $y = \sec x$,

a show that $\frac{d^2y}{dx^2} = 2\sec^3 x - \sec x$

b show that the value of $\frac{d^3y}{dx^3}$ when $x = \frac{\pi}{4}$ is $11\sqrt{2}$.

(P) **6** Given that y is a function of x,

a show that $\frac{d^2}{dx^2}(y^2) = 2y\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2$

b Find an expression, in terms of y, $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ and $\frac{d^3y}{dx^3}$, for $\frac{d^3}{dx^3}(y^2)$.

7 Given that $f(x) = \ln\left(x + \sqrt{1 + x^2}\right)$, show that:

a $\sqrt{1 + x^2}\, f'(x) = 1$ **b** $(1 + x^2) f''(x) + xf'(x) = 0$

c $(1 + x^2) f'''(x) + 3xf''(x) + f'(x) = 0$ **d** Deduce the values of $f'(0)$, $f''(0)$ and $f'''(0)$.

2.3 Maclaurin series

Many functions can be written as an infinite sum of terms of the form ax^n. You may have already encountered series expansions like these:

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \ldots, \ |x| < 1$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \ldots, \ |x| < 1$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \ldots, \ x \in \mathbb{R}$$

Links The first two series expansions shown here are examples of the binomial expansion. ← Pure Year 2, Chapter 4

Example 7

Given that $f(x)$ can be differentiated infinitely many times and that it has a valid series expansion of the form $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots + a_rx^r + \ldots$, where the a_i are all real constants, show that the series expansion must be

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \ldots + \frac{f^{(r)}(0)x^r}{r!} + \ldots$$

Write $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots + a_rx^r + \ldots$

$f(0) = a_0$

The coefficient of a_0 can be found by setting $x = 0$.

Differentiating $f(x)$ gives:

$f'(x) = a_1 + 2a_2x + 3a_3x^2 + \ldots + ra_rx^{r-1} + \ldots$

$f''(x) = 2 \times 1a_2 + 3 \times 2a_3x + \ldots + r(r-1)a_rx^{r-2} + \ldots$

$f'''(x) = 3 \times 2 \times 1a_3 + \ldots + r(r-1)(r-2)a_rx^{r-3} + \ldots$

Successively differentiate with respect to x to obtain $f'(x)$, $f''(x)$ and $f'''(x)$.

Continuing in this way by differentiating r times:

$f^{(r)}(x) = r!a_r +$ terms in powers of x

Evaluate each term at $x = 0$:

$f'(0) = a_1 \Rightarrow a_1 = f'(0)$

$f''(0) = 2!a_2 \Rightarrow a_2 = \frac{f''(0)}{2!}$

$f'''(0) = 3!a_3 \Rightarrow a_3 = \frac{f'''(0)}{3!}$

$f^{(r)}(0) = r!a_r \Rightarrow a_r = \frac{f^{(r)}(0)}{r!}$

Find the coefficients a_1, a_2, $a_3, \ldots, a_r, \ldots$ by substituting $x = 0$ into each result and rearranging.

Therefore $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots + \frac{f^{(r)}(0)}{r!}x^r + \ldots$

Substitute $a_1 = f'(0)$, $a_2 = \frac{f''(0)}{2!}$, $a_3 = \frac{f'''(0)}{3!}, \ldots,$ $a_r = \frac{f^{(r)}(0)}{r!}, \ldots$

In this process, outlined in the worked example above, a polynomial in powers of x is being formed step by step. The process focuses on $x = 0$; substituting $x = 0$ into successive derivatives increases the power of the polynomial. For example, if you stop the process after finding $f'(0)$ the polynomial is linear, $f(0) + f'(0)x$, after $f''(0)$ it is quadratic, $f(0) + f'(0)x + \frac{f''(0)}{2!}x^2$, after $f'''(0)$ it is cubic, $f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3$ and so on.

The above argument assumes that the function can be written in the given form. This is only true if the given series converges. The above reasoning also only holds if the function can be differentiated an infinite number of times, and if $f^{(r)}(0)$ is always finite.

Watch out Not all functions satisfy the condition that $f(0), f'(0), f''(0), \ldots, f^{(r)}(0)$ all have finite values.
For example, when $f(x) = \ln x$, $f'(x) = \frac{1}{x}$ so $f'(0)$ is undefined and therefore does not have a finite value.

- **The Maclaurin series expansion of a function $f(x)$ is given by**

$$\mathbf{f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^{(r)}(0)}{r!}x^r + \ldots}$$

The series is valid provided that $f(0), f'(0), f''(0), \ldots, f^{(r)}(0), \ldots$ all have finite values.

The polynomial $f(0) + f'(0)x$ is a Maclaurin polynomial of degree 1.

The polynomial $f(0) + f'(0)x + \frac{f''(0)}{2!}x^2$ is a Maclaurin polynomial of degree 2.

The polynomial $f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^{(r)}(0)}{r!}x^r$ is a Maclaurin polynomial of degree r.

Even when $f^{(r)}(0)$ exists and is finite for all r, a Maclaurin series expansion is only valid for values of x that give rise to a convergent series. For example, the Maclaurin series of $\frac{1}{1-x}$ is $1 + x + x^2 + x^3 + \ldots$.

But when $x = 2$, the series gives $1 + 2 + 4 + 8 + \ldots$ which does not converge to $\frac{1}{1-2} = -1$.

Note The range of validity for some individual Maclaurin series is given in the formulae booklet. If no range of validity is given in this chapter, you may assume that the expansion is valid for all $x \in \mathbb{R}$.

Example 8

a Express $\ln(1 + x)$ as an infinite series in ascending powers of x.

b Using only the first three terms of the series in part **a**, find estimates for:
i $\ln 1.05$ **ii** $\ln 1.25$ **iii** $\ln 1.8$
Comment on the accuracy of the estimates.

a $f(x) = \ln(1 + x) \Rightarrow f(0) = \ln 1 = 0$

$f'(x) = \frac{1}{1+x} = (1 + x)^{-1} \Rightarrow f'(0) = 1$

$f''(x) = -(1 + x)^{-2} \Rightarrow f''(0) = -1$

$f'''(x) = (-1)(-2)(1 + x)^{-3} \Rightarrow f'''(0) = 2!$

$f^{(r)}(x) = (-1)(-2)(-3)\ldots(-(r - 1))(1 + x)^{-r}$

$\Rightarrow f^{(r)}(0) = (-1)^{r-1}(r - 1)!$

Problem-solving

The term $(-1)^r$ can be used in the general term of **alternating sequences**, in which the terms are alternately positive and negative.

So $\ln(1+x) = 0 + 1x + \frac{-1}{2!}x^2 + \frac{(2!)}{3!}x^3 + \ldots + \frac{(-1)^{r-1}(r-1)!}{r!}x^r + \ldots$

$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \ldots + (-1)^{r-1}\frac{x^r}{r} + \ldots$

b i $\ln 1.05 = 0.05 - \frac{0.05^2}{2} + \frac{0.05^3}{3} - \ldots$

$\approx 0.0487916\ldots$ This is correct to 5 d.p.

ii $\ln 1.25 = 0.25 - \frac{0.25^2}{2} + \frac{0.25^3}{3} - \ldots$

$\approx 0.223958\ldots$ This is correct to 2 d.p.

iii $\ln 1.8 = 0.8 - \frac{0.8^2}{2} + \frac{0.8^3}{3} - \ldots$

$\approx 0.6506666\ldots$ This is not correct to 1 d.p.

Substitute the values for f(0), f′(0), f″(0) etc. into the Maclaurin series for f(x).

Online This expansion is valid for $-1 < x \leqslant 1$. If you use a computer to generate the graphs of the successive Maclaurin polynomials you will see that they converge to the graph of $\ln(1+x)$ between $x = -1$ and $x = 1$, but outside that interval they diverge rapidly. Explore this using GeoGebra.

The further away a value is from $x = 0$, the less accurate the approximation will be and the more terms of the series you need to take to maintain a required degree of accuracy.

Example 9

a Find the first four terms in the Maclaurin series of $\sin x$.

b Using the first two terms of the series find an approximation for $\sin 10°$.

a

$f(x) = \sin x$	$\Rightarrow$	$f(0) = \sin 0 = 0$
$f'(x) = \cos x$	$\Rightarrow$	$f'(0) = \cos 0 = 1$
$f''(x) = -\sin x$	$\Rightarrow$	$f''(0) = -\sin 0 = 0$
$f'''(x) = -\cos x$	$\Rightarrow$	$f'''(0) = -\cos 0 = -1$
$f''''(x) = \sin x$	$\Rightarrow$	$f''''(0) = \sin 0 = 0$

So $\sin x = x + \frac{-1}{3!}x^3 + \frac{1}{5!}x^5 + \frac{-1}{7!}x^7 + \ldots + \frac{(-1)^r}{(2r+1)!}x^{2r+1} + \ldots$

$= x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \ldots$

b $\sin 10° = \sin\frac{\pi}{18} \approx \frac{\pi}{18} - \frac{1}{6}\left(\frac{\pi}{18}\right)^3$

$\approx 0.174532925 - 0.000886096$

≈ 0.173646829

$f^{(n)} = 0$, if n is even, and the cycle of values 0, 1, 0, −1 repeats itself.

This expansion is valid for all values of x.

Watch out x must be in radians in expansions of trigonometric functions.

This estimate is correct to 5 decimal places; even using $\sin x \approx x$, the approximation is correct to 2 d.p.

Exercise 2C

1 Use the formula for the Maclaurin series and differentiation to show that:

a $(1-x)^{-1} = 1 + x + x^2 + \ldots + x^r + \ldots$

b $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \ldots$

Hint The binomial expansions of $(1+x)^n$, where n is fractional or negative and $|x| < 1$, are the Maclaurin series of the function.

2 Use Maclaurin series and differentiation to show that the first three terms in the series expansion of $e^{\sin x}$ are $1 + x + \frac{x^2}{2}$

(P) **3** **a** Show that the Maclaurin series of $\cos x$ is $1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \ldots + (-1)^r \frac{x^{2r}}{(2r)!} + \ldots$

b Using the first three terms of the series, show that it gives a value for $\cos 30°$ correct to 3 decimal places.

Hint This expansion is valid for all values of x.

4 Using the series expansions for e^x and $\ln(1+x)$ respectively, find, correct to 3 decimal places, the values of:

a e **b** $\ln\left(\frac{6}{5}\right)$

5 Use Maclaurin series and differentiation to expand, in ascending powers of x up to and including the term in x^4,

a e^{3x} **b** $\ln(1+2x)$ **c** $\sin^2 x$

(P) **6** Using the addition formula for $\cos(A-B)$ and the series expansions of $\sin x$ and $\cos x$, show that

$$\cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}\left(1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} + \ldots\right)$$

(E) **7** Given that $f(x) = (1-x)^2 \ln(1-x)$,

a show that $f''(x) = 3 + 2\ln(1-x)$ **(2 marks)**

b find the values of $f(0)$, $f'(0)$, $f''(0)$, and $f'''(0)$ **(1 mark)**

c express $(1-x)^2 \ln(1-x)$ in ascending powers of x up to and including the term in x^3. **(3 marks)**

(E/P) **8** **a** Using the series expansions of $\sin x$ and $\cos x$, show that

$$3\sin x - 4x\cos x + x = \tfrac{3}{2}x^3 - \tfrac{17}{120}x^5 + \ldots$$ **(5 marks)**

b Hence, find the limit, as $x \to 0$, of $\frac{3\sin x - 4x\cos x + x}{x^3}$ **(1 mark)**

E **9** Given that $f(x) = \ln \cos x$,

a show that $f'(x) = -\tan x$ **(2 marks)**

b find the values of $f'(0)$, $f''(0)$, $f'''(0)$ and $f''''(0)$ **(1 mark)**

c express $\ln \cos x$ as a series in ascending powers of x up to and including the term in x^4 **(3 marks)**

d show that using the first two terms of the Maclaurin series for $\ln \cos x$, with $x = \frac{\pi}{4}$, gives a value for $\ln 2$ of $\frac{\pi^2}{16}\left(1 + \frac{\pi^2}{96}\right)$. **(2 marks)**

E/P **10** Show that the Maclaurin series for $\tan x$, as far as the term in x^5, is $x + \frac{1}{3}x^3 + \frac{2}{15}x^5$. **(5 marks)**

Challenge

The **ratio test** is a sufficient condition for the convergence of an infinite series. It says that a series $\sum_{r=1}^{\infty} a_r$ converges if $\lim_{r\to\infty}\left|\frac{a_{r+1}}{a_r}\right| < 1$, and diverges if $\lim_{r\to\infty}\left|\frac{a_{r+1}}{a_r}\right| > 1$.

Use the ratio test to show that

a the Maclaurin series expansion of e^x converges for all $x \in \mathbb{R}$

b the Maclaurin series expansion of $\ln(1 + x)$ converges for $-1 < x < 1$, and diverges for $x > 1$.

Problem-solving

If $\lim_{r\to\infty}\left|\frac{a_{r+1}}{a_r}\right| = 1$ or does not exist then the ratio test is inconclusive.

2.4 Series expansions of compound functions

You can find the series expansions of compound functions using known Maclaurin series. In the last exercise you found the Maclaurin series of simple compound functions, such as e^{3x} and $\ln(1 + 2x)$. However, the resulting series could also be found by replacing x by $3x$ or x by $2x$ in the known expansions of e^x and $\ln(1 + x)$ respectively. When successive derivatives of a compound function are more difficult, or when there are products of functions involved, it is often possible to use one of the standard results.

■ **The following Maclaurin series expansions are given in the formulae booklet:**

- $\mathbf{e^x = 1 + x + \frac{x^2}{2!} + \ldots + \frac{x^r}{r!} + \ldots}$ **for all x**
- $\mathbf{\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + (-1)^{r+1}\frac{x^r}{r} + \ldots}$ $\mathbf{-1 < x \leqslant 1}$
- $\mathbf{\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots + (-1)^r\frac{x^{2r+1}}{(2r+1)!} + \ldots}$ **for all x**
- $\mathbf{\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots + (-1)^r\frac{x^{2r}}{(2r)!} + \ldots}$ **for all x**
- $\mathbf{\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots + (-1)^r\frac{x^{2r+1}}{2r+1} + \ldots}$ $\mathbf{-1 \leqslant x \leqslant 1}$

Example 10

Write down the first four non-zero terms in the series expansion, in ascending powers of x, of $\cos(2x^2)$.

$$\cos(2x^2) = 1 - \frac{(2x^2)^2}{2!} + \frac{(2x^2)^4}{4!} - \frac{(2x^2)^6}{6!} + \ldots$$

$$= 1 - 2x^4 + \frac{2}{3}x^8 - \frac{4}{45}x^{12} + \ldots$$

Substitute $2x^2$ for x in the above series for $\cos x$.

Watch out Make sure you simplify the coefficients as much as possible.

Example 11

Find the first three non-zero terms in the series expansion of $\ln\left(\frac{\sqrt{1+2x}}{1-3x}\right)$, and state the values of x for which the expansion is valid.

$$\ln\left(\frac{\sqrt{(1+2x)}}{1-3x}\right) = \ln\sqrt{1+2x} - \ln(1-3x)$$

Using $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$

$$= \tfrac{1}{2}\ln(1+2x) - \ln(1-3x)$$

Using $\ln a^{\frac{1}{2}} = \frac{1}{2}\ln a$

$$\tfrac{1}{2}\ln(1+2x) = \tfrac{1}{2}\left(2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} - \ldots\right), \quad -1 < 2x \leqslant 1$$

Substitute $2x$ for x in the expansion of $\ln(1+x)$.

$$= x - x^2 + \tfrac{4}{3}x^3 - \ldots \qquad -\tfrac{1}{2} < x \leqslant \tfrac{1}{2}$$

Problem-solving

You are substituting $2x$ into the series expansion of $\ln(1+x)$, so the series is now only valid for $-1 < 2x \leqslant 1$, or $-\frac{1}{2} < x \leqslant \frac{1}{2}$

$$\ln(1-3x) = (-3x) - \frac{(-3x)^2}{2} + \frac{(-3x)^3}{3} - \ldots, \quad -1 < -3x \leqslant 1$$

$$= -3x - \tfrac{9}{2}x^2 - 9x^3 - \ldots \qquad -\tfrac{1}{3} \leqslant x < \tfrac{1}{3}$$

Substitute $-3x$ for x in the expansion of $\ln(1+x)$.

$$\text{So } \ln\frac{\sqrt{1+2x}}{1-3x} = (x - x^2 + \tfrac{4}{3}x^3 - \ldots)$$

$$- (-3x - \tfrac{9}{2}x^2 - 9x^3 - \ldots), \quad -\tfrac{1}{3} \leqslant x < \tfrac{1}{3}$$

$$= 4x + \tfrac{7}{2}x^2 + \tfrac{31}{3}x^3 + \ldots, \qquad -\tfrac{1}{3} \leqslant x < \tfrac{1}{3}$$

You need both intervals to be satisfied. This is the case for $-\frac{1}{3} \leqslant x < \frac{1}{3}$

Example 12

Given that terms in x^n with $n > 4$ may be neglected, use the series expansions for e^x and $\sin x$ to show that

$$e^{\sin x} \approx 1 + x + \frac{x^2}{2} - \frac{x^4}{8}$$

$\sin x = x - \frac{x^3}{3!} + \ldots$

Only two terms are used as the next term is kx^5.

So $e^{\sin x} = e^{\left(x - \frac{x^3}{3!} + \ldots\right)}$

$= e^x \times e^{-\frac{x^3}{6}} \times \ldots$

Use $e^{a-b} = e^a \times e^{-b}$

$= \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \ldots\right)\left(1 + \left(-\frac{x^3}{6}\right) + \ldots\right) \ldots$

Substitute $-\frac{x^3}{6}$ for x in the expansion of e^x.

$= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} - \frac{x^3}{6} - \frac{x^4}{6} + \ldots$

$\approx 1 + x + \frac{x^2}{2} - \frac{x^4}{8}$

Simplify as much as possible.

Exercise 2D

(P) **1** Use the series expansions of e^x, $\ln(1 + x)$ and $\sin x$ to expand the following functions as far as the fourth non-zero term. In each case state the values of x for which the expansion is valid.

a $\frac{1}{e^x}$ **b** $\frac{e^{2x} \times e^{3x}}{e^x}$

c e^{1+x} **d** $\ln(1 - x)$

e $\sin\left(\frac{x}{2}\right)$ **f** $\ln(2 + 3x)$

Hint For part **f**, write $2 + 3x$ as $2\left(1 + \frac{3x}{2}\right)$.

(E/P) **2 a** Using the Maclaurin series of $\ln(1 + x)$, show that

$$\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \ldots\right), -1 < x < 1$$ **(4 marks)**

b Deduce the series expansion for $\ln\sqrt{\frac{1+x}{1-x}}$, $-1 < x < 1$ **(2 marks)**

c By choosing a suitable value of x, and using only the first three terms of the series from part **a**, find an approximation for $\ln\left(\frac{2}{3}\right)$, giving your answer to 4 decimal places. **(2 marks)**

d Show that the first three terms of your series from part **b**, with $x = \frac{3}{5}$, give an approximation for $\ln 2$, which is correct to 2 decimal places. **(2 marks)**

(E/P) **3** Show that, for small values of x, $e^{2x} - e^{-x} \approx 3x + \frac{3}{2}x^2$. **(4 marks)**

(E/P) **4 a** Show that $3x \sin 2x - \cos 3x = -1 + \frac{21}{2}x^2 - \frac{59}{8}x^4 - \ldots$ **(5 marks)**

b Hence find the $\lim_{x \to 0}\left(\frac{3x \sin 2x - \cos 3x + 1}{x^2}\right)$ **(1 mark)**

(P) **5** Find the series expansions, up to and including the term in x^4, of:

a $\ln(1 + x - 2x^2)$

b $\ln(9 + 6x + x^2)$

Note Factorise the quadratic first.

and in each case give the range of values of x for which the expansion is valid.

(E/P) **6 a** Write down the series expansion of $\cos 2x$ in ascending powers of x, up to and including the term in x^8. **(3 marks)**

b Hence, or otherwise, find the first four non-zero terms in the series expansion for $\sin^2 x$. **(3 marks)**

(E/P) **7** Show that the first two non-zero terms of the series expansion, in ascending powers of x, of $\ln(1 + x) + (x - 1)(e^x - 1)$ are px^3 and qx^4, where p and q are constants to be found. **(6 marks)**

(E/P) **8 a** By considering the product of the series expansions of $\sin x$ and $(1 - x)^{-2}$, expand $\dfrac{\sin x}{(1 - x)^2}$ in ascending powers of x as far as the term in x^4. **(6 marks)**

b Deduce the gradient of the tangent, at the origin, to the curve with equation $y = \dfrac{\sin x}{(1 - x)^2}$ **(3 marks)**

(P) **9** Use the Maclaurin series, together with a suitable substitution, to show that:

a $(1 - 3x)\ln(1 + 2x) = 2x - 8x^2 + \frac{26}{3}x^3 - 12x^4 + \ldots$

b $e^{2x}\sin x = x + 2x^2 + \frac{11}{6}x^3 + x^4 + \ldots$

c $\sqrt{1 + x^2}e^{-x} = 1 - x + x^2 - \frac{2}{3}x^3 + \frac{1}{6}x^4 + \ldots$

(E/P) **10 a** Write down the first five non-zero terms in the series expansions of $e^{-\frac{x^2}{2}}$ **(3 marks)**

b Using your result from part **a**, find an approximate value for $\displaystyle\int_{-1}^{1} e^{-\frac{x^2}{2}}\,dx$, giving your answer to 3 decimal places. **(3 marks)**

(E/P) **11 a** Show that $e^{px}\sin 3x = 3x + 3px^2 + \dfrac{3(p^2 - 3)}{2}x^3 + \ldots$ where p is a constant. **(5 marks)**

b Given that the first non-zero term in the expansion, in ascending powers of x, of $e^{px}\sin 3x + \ln(1 + qx) - x$ is kx^3, where k is a constant, find the values of p, q and k. **(4 marks)**

(E/P) **12** $f(x) = e^{x - \ln x}\sin x$, $x > 0$

a Show that, if x is sufficiently small, x^4 and higher powers of x may be neglected,

$$f(x) \approx 1 + x + \frac{x^2}{3}$$ **(5 marks)**

b Show that using $x = 0.1$ in the result from part **a** gives an approximation for $f(0.1)$ which is correct to 6 significant figures. **(2 marks)**

E/P **13** $y = \sin 2x - \cos 2x$

a Show that $\frac{d^4y}{dx^4} = 16y$ **(4 marks)**

b Find the first five terms of the Maclaurin series for y, giving each coefficient in its simplest form. **(4 marks)**

Challenge

The Lorentz factor of a moving object, λ, is given by the formula

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

where $\beta = \frac{v}{c}$ is the ratio of v, the speed of the object, to c, the speed of light (3×10^8 m s^{-1}).

a Find the Maclaurin series expansion of $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ in ascending powers of β up to the term in β^4

The theory of special relativity predicts that a period of time observed as T within a stationary frame of reference will be observed as a period of time $\frac{T}{\gamma}$ in a moving frame of reference.

A spaceship travels from Earth to a planet 4.2 light years away. To an observer on Earth, the journey appears to take 20 years.

b Use your answer to part **a** to estimate the observed journey time for a person on the spaceship.

c Calculate the percentage error in your estimate.

d Comment on whether your approximation would be more or less accurate if the spaceship was travelling at three times the speed.

Note A light year is the distance light travels in one year.

Mixed exercise 2

E/P **1 a** Express $\frac{2}{(r+2)(r+4)}$ in partial fractions. **(1 mark)**

b Hence show that $\sum_{r=1}^{n} \frac{2}{(r+2)(r+4)} = \frac{7n^2+25n}{12(n+3)(n+4)}$ **(5 marks)**

E/P **2 a** Express $\frac{4}{(4r-1)(4r+3)}$ in partial fractions. **(2 marks)**

b Using your answer to part **a** and the method of differences, show that

$$\sum_{r=1}^{n} \frac{4}{(4r-1)(4r+3)} = \frac{4n}{3(4n+3)}$$ **(3 marks)**

c Evaluate $\sum_{r=100}^{200} \frac{4}{(4r-1)(4r+3)}$ giving your answer to 3 significant figures. **(2 marks)**

(E) **3 a** Show that $(r+1)^3-(r-1)^3=6r^2+2$. **(2 marks)**

b Using the result from part **a** and the method of differences, show that

$$\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$$ **(5 marks)**

(E/P) **4** Prove that $\sum_{r=1}^{n} \frac{4}{(r+1)(r+3)} = \frac{n(an+b)}{3(n+2)(n+3)}$, where a and b are constants to be found. **(5 marks)**

(E/P) **5** Prove that $\sum_{r=n}^{2n}((r+1)^3-(r-1)^3) = an^3+bn^2+cn+d$, where a, b, c and d are constants to be found. **(5 marks)**

6 a Given that $y=e^{1-2x}$, find an expression, in terms of y, for $\frac{d^n y}{dx^n}$

b Hence show that $\frac{d^8 y}{dx^8}$ at $x=\ln 32$ is $\frac{e}{4}$

7 a For the function $f(x)=\ln(1+e^x)$, find the values of $f'(0)$ and $f''(0)$.

b Show that $f'''(0)=0$.

c Find the series expansion of $\ln(1+e^x)$, in ascending powers of x up to and including the term in x^2.

(E/P) **8 a** Write down the Maclaurin series of $\cos 4x$ in ascending powers of x, up to and including the term in x^6. **(3 marks)**

b Hence, or otherwise, show that the first three non-zero terms in the series expansion of $\sin^2 2x$ are $4x^2-\frac{16}{3}x^4+\frac{128}{45}x^6$. **(3 marks)**

(E/P) **9** Given that terms in x^5 and higher powers may be neglected, use the Maclaurin series for e^x and $\cos x$, to show that $e^{\cos x} \approx e\left(1-\frac{x^2}{2}+\frac{x^4}{6}\right)$. **(5 marks)**

(E/P) **10** Given that $|2x|<1$, find the first two non-zero terms in the series expansion of $\ln((1+x)^2(1-2x))$ in ascending powers of x. **(5 marks)**

(E/P) **11** Use differentiation and Maclaurin series, to express $\ln(\sec x+\tan x)$ as a series in ascending powers of x up to and including the term in x^3. **(5 marks)**

(P) **12** Show that the results of differentiating the standard series expansions of e^x, $\sin x$ and $\cos x$ agree with the following:

a $\frac{d}{dx}(e^x)=e^x$ **b** $\frac{d}{dx}(\sin x)=\cos x$ **c** $\frac{d}{dx}(\cos x)=-\sin x$

(E) **13 a** Given that $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots$, show that $\sec x = 1 + \frac{x^2}{2} + \frac{5}{24}x^4 + \ldots$. **(4 marks)**

b Using the result found in part **a**, and given that $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$, find the first three non-zero terms in the series expansion, in ascending powers of x, for $\tan x$. **(4 marks)**

(E/P) **14** By using the series expansions of e^x and $\cos x$, or otherwise, find the expansion of $e^x \cos 3x$ in ascending powers of x up to and including the term in x^3. **(5 marks)**

(E/P) **15** Find the first three derivatives of $(1 + x)^2 \ln(1 + x)$. Hence, or otherwise, find the expansion of $(1 + x)^2 \ln(1 + x)$ in ascending powers of x up to and including the term in x^3. **(5 marks)**

(E/P) **16 a** Expand $\ln(1 + \sin x)$ in ascending powers of x up to and including the term in x^4. **(4 marks)**

b Hence find an approximation for $\int_0^{\frac{\pi}{6}} \ln(1 + \sin x)\,dx$ giving your answer to 3 decimal places. **(3 marks)**

(E/P) **17 a** Using the first two terms, $x + \frac{x^3}{3}$, in the expansion of $\tan x$, show that

$$e^{\tan x} = 1 + x + \frac{x^2}{2} + \frac{x^3}{2} + \ldots$$ **(3 marks)**

b Deduce the first four terms in the series expansion of $e^{-\tan x}$, in ascending powers of x. **(3 marks)**

(P) **18 a** Using Maclaurin series, and differentiation, show that $\ln \cos x = -\frac{x^2}{2} - \frac{x^4}{12} + \ldots$

b Using $\cos x = 2\cos^2\left(\frac{x}{2}\right) - 1$, and the result in part **a**, show that

$$\ln(1 + \cos x) = \ln 2 - \frac{x^2}{4} - \frac{x^4}{96} + \ldots$$

(E/P) **19** $y = e^{3x} - e^{-3x}$

a Show that $\frac{d^4y}{dx^4} = 81y$. **(4 marks)**

b Find the first three non-zero terms of the Maclaurin series for y, giving each coefficient in its simplest form. **(3 marks)**

c Find an expression for the nth non-zero term of the Maclaurin series for y. **(2 marks)**

Challenge

Given that the Maclaurin series of e^x is valid for all $x \in \mathbb{C}$, show, using series expansions, that $e^{ix} = \cos x + i \sin x$.